SEFI: Conceptual and Foundational Aspects of a Unified Geometric Framework

Jason Dutton

August 20, 2026

Abstract

SEFI (Structured Emergent Field Integration) proposes that physical, biological, and cognitive systems share a common geometric substrate. In this conceptual version of the SEFI framework, we emphasize interpretational principles, the motivations for unification, and the philosophical implications of treating diverse phenomena as manifestations of manifold geometry. We present the canonical SEFI warp-metric, operator set, SEFI:DNA biological model, and unified evolution equation, focusing on their conceptual coherence and cross-domain applicability. We discuss how SEFI reframes questions about emergence, causation, and structure, and outline pathways for empirical and theoretical evaluation.

1 Introduction: Why geometry unifies

The search for unification in physics traditionally aims to combine interactions or fields into a single mathematical structure. SEFI takes a broader view: geometry itself may be the common language of physical, biological, and cognitive systems. This perspective aligns with long-standing ideas in foundational physics, but extends them by proposing a minimal operator set capable of describing structure, change, and coherence across domains.

This paper presents the conceptual backbone of SEFI, emphasizing interpretational clarity rather than technical detail. The mathematical definitions are identical to those in the technical version, but the framing is philosophical and structural.

2 Geometric assumptions and interpretational stance

SEFI assumes:

- Systems are represented as manifolds or submanifolds.
- Geometry encodes both structure and dynamics.
- Operators represent generalized "forces" or "tendencies" acting on fields.
- Warp fields encode contextual or environmental modulation.

These assumptions are not empirical claims but conceptual commitments. They provide a unified language for describing diverse systems.

3 Canonical SEFI warp-metric

The warp-metric formalism expresses how contextual modulation alters geometry. Given a reference metric $g_{\mu\nu}$ and warp field $W(x)$, the canonical warp correction is

$$F_{\mu\nu}[W] = \alpha W g_{\mu\nu} + \beta \nabla_\mu \nabla_\nu W,$$

and the SEFI metric is

$$\tilde{g}_{\mu\nu} = g_{\mu\nu} + F_{\mu\nu}[W].$$

Conceptually, the warp-metric encodes how systems "bend" under internal or external influences. The second-derivative term represents structural sensitivity.

4 Canonical operator set

SEFI posits four operators:

- **Warp operator** $\hat{W}$: contextual modulation.
- **Tension operator** $\hat{T}$: flow, routing, or stress.
- **Coherence operator** $\hat{C}$: persistence of structure.
- **Correction operator** $\hat{R}$: restoration or equilibration.

These operators are not tied to specific physics; they represent conceptual categories of change applicable across domains.

5 SEFI:DNA as a conceptual biological model

Biological systems are modeled as submanifolds $\mathcal{B}$ with state field $\Psi(x, t)$. The canonical evolution equation

$$\partial_t \Psi = \hat{R}\Psi + \hat{D}\Psi - \hat{A}\Psi$$

captures restoration, duplication, and aging. Conceptually:

- $\hat{R}$: healing and homeostasis.
- $\hat{D}$: growth and reproduction.
- $\hat{A}$: curvature-induced fatigue.

This model treats biological processes as geometric transformations.

6 Unified manifold evolution

SEFI cosmology generalizes geometric flow:

$$\partial_t \tilde{g}_{\mu\nu} = -2\kappa_G \left(R_{\mu\nu} - \tfrac{1}{2} R \tilde{g}_{\mu\nu}\right) + S_{\mu\nu}[W, T, C].$$

Conceptually, this expresses how global structure evolves under warp, tension, and coherence influences. It reframes cosmology as a balance of geometric tendencies rather than purely stress-energy dynamics.

7 Interpretational implications

SEFI suggests:

- Emergence may be geometric rather than statistical.
- Biological and cognitive coherence may be curvature phenomena.
- Contextual modulation (warp) may unify environmental and internal effects.
- Cosmology may be governed by generalized geometric flows.

These implications are conceptual and require empirical grounding.

8 Conclusion

This version of SEFI emphasizes conceptual clarity and interpretational coherence. The mathematical structures are minimal but sufficient to express a unified geometric perspective across domains. Future work will refine the technical details and explore empirical consequences.

References